

Trends of Friends – Time dynamics of Surface- and Deep- level traits in friendship formation and maintenance

Daniel Cowen^{a,*}, Tobias Stark^b, Vincenz Frey^a, Andreas Flache^a

^a Department of Sociology, University of Groningen, Netherlands

^b Department of Interdisciplinary Social Science, Utrecht University, Netherlands

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ABSTRACT

People become friends with one another primarily due to things they have in common, like shared demographic characteristics or shared interests. But on what similarities are people becoming friends at different stages of knowing one another? To study this, we use a longitudinal dataset that followed a cohort of students of one study programme at a Swiss technical university. We model how demographic traits of nationality and gender, and less observable interest-related traits of being social, a partygoer, a smart and hard-working student, contribute to friendships. Using Stochastic Actor- Oriented Models, we find a baseline level of both demographic and interest-related friendship homophily, indicating that there are differing reasons behind friendships. We also find that homophily based on demographic traits diminishes over time when students get to know those in their study cohort better. This suggest that homophily on observable traits is mainly relevant when people first meet but becomes less important over time.

1. Introduction

Similarity begets friendship. In 370 BCE, Plato's Phaedrus codified this quote, and it has since developed into the cardinal idea of homophily. Research on homophily has shown that people are drawn to and entertained by others similar to themselves: Whether it is the country one grew up in or activities we love doing, we are drawn to those with whom we share our culture and can discuss intricate details of the things we love (McPherson, Smith-Lovin, and Cook, 2001). It is hard to dispute that similarity draws us together; however, which similarity does so is less clear. This question arises in settings where people come together who did not know each other before, as for example among freshmen in university. On which commonalities do people in such a setting create friendships with one another, and how does the importance of similarity in different aspects change over time as one learns more about one's friends and other people around?

This paper addresses this question by studying the evolution of a friendship network composed of individuals with diverse demographic and interest/personality-related traits. We provide a systematic investigation of the interplay of demographic and non-demographic traits over time, observing friendship development in a new university cohort where most students meet each other for the first time. This allows us to

observe how individuals develop friendships with different people as they learn more and more about one another, with waves collected more frequently and in shorter intervals than other studies in the same domain (Block and Grund, 2014; Leszczensky and Pink, 2019). We argue that early on in getting to know one another, friendships will form based on demographic homophily, leading to gender and nationality-based clustering. However, as time passes, people may realise that the shared demographic group membership does not always align with shared values and interests, making these friendships less rewarding. They may also realise they are more similar in these values, interests and personality traits to some people with whom they do not share demographic characteristics, making them attractive for friendship formation. We thus argue that, over time, friendship formation based on similar interests and values can promote integration across demographic lines. To empirically test these processes and their implications for the demographic clustering of friendship networks, this paper examines how the role of demographics in friendship formation changes over time. We also test if non-demographic traits become more relevant for friendships over time and how this affects the demographic clustering of the network.

We use the Swiss Student Life study (Elmer et al., 2022; Vörös et al., 2021) to longitudinally disentangle how friendships develop in a closed

* Correspondence to: University of Groningen, B0111, Bouman Building, Grote Rozenstraat 31, Groningen 9712 TG, Netherlands.

E-mail address: d.r.cowen@rug.nl (D. Cowen).

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cohort of university students all following the same courses in the same study program ($N = 593$). With five survey waves administered over their first study year, we can observe micro-processes in developing friendships among students unfamiliar with one another. These students come into contact randomly as the year progresses or through events that make certain traits salient – such as an end-of-year exam they are required to pass to continue to the next year of their studies. To observe this, variables based on self-reported demographic characteristics and personality traits as well as students perceived personality traits and hobbies are operationalised. We use Stochastic Actor Oriented Models (Snijders, 2017; Snijders et al., 2024) to disentangle the different forms of homophily we are interested in from other simultaneous processes of friendship formation. Interactions between homophily effects and time enable us to model how their relevance for friendship formation changes as the year passes

1.1. Prior research

Many studies show that shared socio-demographic characteristics foster friendship formation (Aiello et al., 2012), be it similarity on race (Shrum, Cheek, and Hunter, 1988), ethnicity (Stark and Flache, 2012; Syed and Juan, 2012; Zhao, 2023) or gender (Smith, McPherson, and Smith-Lovin, 2014). On an individual level, everyone has various group memberships, interests, and values, often not easily visible, some related to their socio-demographic characteristics (Leszczensky and Pink, 2019; Smith et al., 2014), while others are not. When friendships are created in a context where people do not initially know one another, relationships established based on demographic similarity can be expected. Demographic characteristics are often easily recognisable and demographic similarity tends to go hand in hand with similarities in underlying cultural dimensions. In contrast, with enough time to learn about one another, some might reach across demographic lines to find new friends with similar traits more important to them than demographics, such as similar interests or personality traits.

Lazarsfeld and Merton (1954) coined the concept of value homophily to describe the tendency to befriend others with similar traits (values, interests and personalities) not necessarily aligned with demographics (what they called status homophily) and that are not instantly apparent. Instead, these traits are recognized only after others had sufficient occasions to express particular values. Lazarsfeld and Merton's argument of value homophily is that shared values and interests are a more robust basis for friendships – so much so that one might favour individuals from other demographic groups with whom one shares common interests instead of demographically similar individuals with whom one shares few interests. Nevertheless, despite decades of research into demographic and value homophily (Bhargava et al., 2022; McPherson et al., 2001), their interplay over time is still poorly understood.

This reasoning also implies that value homophily may be difficult to detect based on self-reports alone. Particularly in early stages of knowing one another (Selfhout et al., 2009), self-reported interests or personality traits may not be widely known throughout a population, leading biased estimation of homophily (Montoya, Horton, and Kirchner, 2008). For example, a student may, in fact, be the smartest in their cohort, but if this is not well known, peers may befriend others who they perceive to be the smartest instead. Empirically, this discrepancy between self-reports and other students' perceptions is observed in disruptive behaviour (Branson and Cornell, 2009; Henry et al., 2006) and personality characteristics (Selfhout et al., 2009). To address this challenge, we use both students' perceptions of individuals interest traits through peer-nominations, as well as self-reports of personality traits in separate models to look at the different sources of characteristics driving value homophily.

Most research that looks at less visible, value-driven characteristics underlying homophily focuses on specific characteristics in isolation, such as delinquency and associated actions like smoking (Kandel, 1978; Urberg, Degirmencioglu, and Tolson, 1998) playing video games

(Eklund and Roman, 2017), being an extravert (Van Zalk et al., 2020), or enjoying sports (Dalen and Seippel, 2021). While there is work done looking at homophily on multiple less visible dimensions, this is often done without testing for changes in homophily over time (Beek et al., 2024; Schaefer et al., 2011; Selfhout et al., 2010; S. Smith et al., 2014; Stark and Flache, 2012), often over years rather than months (Steglich, Snijders, and West, 2006), and with fewer observation times in a year (Schaefer et al., 2010; Snijders, Lomi, and Torló 2013), all without looking at effects between waves - leading to a lack of opportunities to capture more nuanced processes between waves rather than throughout all waves. Resultingly, there is insufficient understanding of how the dynamics between demographic characteristics and these values affect friendship over time, particularly in contexts where the nodes have no prior knowledge of those around them when first observed.

2. Theory and hypothesis

2.1. Homophily on surface-level and deep-level characteristics

The well-established homophily effect on easily observable demographic characteristics such as race and ethnicity is often a product of individuals from the same demographic group possessing similar cultural values or related preferences. Factors such as interests and hobbies correlating with demographic characteristics (Smith et al., 2014; Stark and Flache, 2012), ease of communication (Rogers and Bhowmik, 1970), shared norms and behaviour (Bhargava et al., 2022), and trust (Ahlf et al., 2019) aggregate to create cultural interests. These, in turn, lend themselves towards an in-group preference for one another on an individual level, in line with social identity theory (Reicher, Spears, and Halsam, 2010). For example, individuals prefer others from the same country (McPherson et al., 2001) – not necessarily due to being from the same country. Instead, being from a specific country might correlate with a particular set of cultural preferences – such as ease of communication when speaking the same language, preferences for favourite sports and food, shared understanding of national norms such as etiquette and hierarchy, as well as accumulated trust due to positive prior and reinforcing experiences with one's own in-group. These demographic characteristics are often used as an initial dimension to befriend one another due to perceived associated correlated characteristics, known as ethnic by-products (Smith et al., 2014; Stark and Flache, 2012), where demographics act as a signal of specific cultural preferences related to one's demographic group. In this paper, we define these demographic traits as surface-level characteristics because they are easily observable (Phillips, Northcraft, and Neale, 2006; Phillips and Loyd, 2006). The low-cost nature of this friendship selection mechanism, regarding time investment, is beneficial, as other assumed shared interests are expected to be correlated with the demographic trait (Leszczensky and Pink, 2015; Stark and Flache, 2012). Thus, we propose:

H1. Similarity on surface-level (demographic) characteristics drives friendship formation.

Everyone possesses cultural interests and associated by-products that initially make friendships based on surface-level demographic characteristics more likely. However, only some people hold these interests and preferences to the same extent. Further, some people may not prefer their assumed demographic by-products at all but rather unrelated features, such as certain hobbies and interests (Byrne and Nelson, 1965) or similar personality traits (Bradley et al., 2013; Cuperman and Ickes, 2009; Van Zalk et al., 2020). For example, an individual may have been born in the Netherlands but has no interest in Dutch cuisine and does not watch Dutch TV shows. Instead, this individual may want to befriend others who enjoy the same hobbies as they do, such as playing video games (Eklund and Roman, 2017), going out to party (Ingram and Morris, 2007), being extraverted (Van Zalk et al., 2020), or being a diligent student (Brouwer et al., 2018; Smirnov and Thurner, 2016)

Should friendships be formed based on these characteristics rather than demographic characteristics, then while homophily is still present, it is no longer primarily driven by surface-level traits. Instead, it is driven by preferences in hobbies, personality traits, or values, which we term deep-level characteristics (Phillips and Loyd, 2006; Phillips et al., 2006).

H2. Similarity in deep-level characteristics (non-demographic) drives friendship formation.

2.2. The role of surface-level and deep-level similarity over time

Homophily is not an isolated phenomenon but rather a continuous process subjected to various contexts and events that might change what someone deems necessary to base friendship upon. In this, we assume that the friends made when first exposed to a new cohort of people are based on different mechanisms than later in the same cohort. This can especially be the case with demographic homophily, which might fluctuate over time¹ (Block and Grund, 2014; Boda et al. 2020). Deep-level factors may not be salient at an early stage of familiarisation, making it a challenge to find others with similar values/interests. In contrast, surface-level characteristics such as nationality and gender are salient from the outset, making them relevant for friendship formation when no other information is present. However, this reliance on surface-level characteristics will decrease over time as people find others similar to them on other dimensions. This might be either randomly or driven by specific events that more efficiently draw out these deep-level characteristics. In our case, such an event could be the exam at the end of the first year of studies, making diligent students looking for similar others, or introduction events at the beginning of the year that drive people to find others whom they can socialise/party with. Thus, we propose:

H3a. Over time, shared surface-level characteristics become weaker predictors of friendship formation.

H3b. Over time, shared deep-level characteristics become stronger predictors of friendship formation.

2.3. Homophily of surface-level characteristics over time

Importantly, while individual preferences solely drive friendship selection behaviour at the micro- level, they may have profound macro-level implications. Individuals' search for friends with characteristics unrelated to demographics offers an opportunity for macro-social integration based on interests and preferences for friends outside of demographically driven social group categories (Blau and Schwartz, 1984), in our case, nationality and gender. Given that, in previous hypothesis, we expect diverse interests and personality traits to dictate friendship decisions, we think that these dynamics will override grouping based on nationality or gender. Thus, our hypothesized increased reliance on non-correlated deep-level factors over time should negate the formation of demographically separated friendship groups early in friendship for all demographic subgroups studied. Resultantly, this should reduce demographic homophily across all surface-level characteristics.

Of interest is also the macro-level effect of these dyadic level choices – particularly whether changes in the homophily preferences lead to more cross-demographic friendships in general, or whether specific demographic groups tend to be less homophilic than others. Prior research has shown how degree and group sizes (Sajjadi et al., 2024), age (Goel, Sharma, and Aasa, 2021), and gender (Pignolet, Schmid, and Seelisch, 2024) may be associated with different homophily preferences.

¹ This is based on an intervention study with the same data, where the students were put into gender-homogenous or heterogeneous groups. They were then studied to see how long friendships remained based on the people in this group, finding that groups of mostly cross-gender students remained friends for longer than same-gender groups.

To assess potential macro-level effect, we compare the level of homophily within different groups over time to random mixing.

H4. Over time, the friendship network will have lower levels of homophilic clustering based on demographics.

3. Data and methodology

3.1. Data

We use data from the Swiss Student Life study,² a longitudinal study on friendships and study networks conducted among a cohort of students entering a Swiss technical university in the academic years from 2017 to 2020 ($n = 652$) (Vörös et al. 2021). We focus on the first five waves of the cohort, with 593 respondents who nominated at least one student or were nominated at least once within the first year of their studies. These students all enter university at the same time and follow the same first year courses. The exact dates for each collection can be found in Fig. 1. This data has previously been used to examine how membership in study and friendship groups affects academic performance (Stadtfeld et al. 2019). Students must pass an exam at the end of their first year, determining whether they will continue to the program's second year. Approximately 54 % of the students pass on their first attempt, with the remaining 46 % having to repeat the year or drop out. This is why we restrict our analysis to the first year. Network information was collected using name generators that allowed respondents to list a maximum of 20 alters. For the current study, we use the network about friendship relations. The gender composition of the data is approximately 13 % females, 76 % males, and 11 % did not disclose their gender. For the nationality composition, about 67 % are Swiss, 26 % are international, and 7 % are not disclosed.

Five detailed online questionnaires were administered in their first year, with an average of two months between each questionnaire. The students were recruited based on lists from the university's student services. Given the flexibility of the study program and the integration of students who failed the exam in previous cohorts, the composition of the network throughout the year varied depending on the student's academic trajectory.

3.2. Measures

A breakdown of each variable used can be found in S1 in the online supplement. Questions for self-reported traits can be found in S2.

3.2.1. Surface-level traits

Surface-level characteristics pertain to country of birth, coded as *Swiss* (1) and *Other* (0), and gender, coded as *Male* (1) and *Female* (0). There were also open questions for 'other' categories on these constructs, but we focus here on the dichotomy local vs. international student due to many specific countries of origin, and there were no transgender, intersex, or non-binary respondents in the data. For robustness checks, we run models on more granular country definitions (Swiss, Germanic, English, and Other) that produced similar results. See S3 in the online supplement.

3.2.2. Deep-level traits

3.2.2.1. Nomination based perceptions. To measure deep-level characteristics, we use questions that asked respondents about the perceptions

² This data was collected by the Swiss Student Life team at ETH Zurich. Due to sensitivity and restrictions on the dataset by requesting, it cannot be published publicly with this paper. The code for the models can be found in the replication package and more information on the dataset and how to request it can be found in the codebook (Elmer et al. 2022)



Fig. 1. Friendship networks in the first year of the study program. Color indicate students' gender (brown = male, light blue = female, grey = missing information on gender). Isolates in the figures are students who neither nominated nor were nominated by others in the cohort in a particular wave but nominated or were nominated in another wave. These students may not (yet) know anyone, have friends exclusively outside of the program, or, in later waves, have dropped out of the program.

of other students in their cohort, asking them to name five students who they perceive as holding a particular trait. The questions asked are “Who is particularly smart among your fellow students?” for being *clever*, “Who of your fellow students takes their studies very seriously?” for *serious*, “Who of your fellow students parties a lot?” for *partygoers* and “Which of your fellow students would be most likely to start social activities?” for *social*. For each student, we counted the number of nominations they received (indegree). For each wave, this results in the overall perceptions of how strongly each student possesses these traits in the eyes of their cohort. In this, similarity in the number of times a student is nominated in these categories will be established to predict who befriends whom. Distributions of the number of nominations per trait can be found in S4 in the online supplement.

3.2.2.2. Self-reported big-5 personality traits. To see whether there are discrepancies between self-reported traits and peer perceptions of other students, we also run models on self-reports on Big 5 Personality Traits (McCrae and Costa, 1987) as an alternative operationalisation of deep-level traits. These are derived from a 9-item scale per construct asked of the students relating to these concepts, resulting in 45 total questions relating to these personality traits. Each question offers a 5-point Likert scale with the degree to which they agree that they hold a particular trait or behaviour within each construct. From this, we derive averages per construct - resulting in a value between 1 and 5 for each factor (Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism) for each student. These questions can be found in S2 in the

[supplementary materials](#).

3.2.3. Controls

Our models account for well-established network phenomena in friendship formation processes, such as triads (Block, 2015; Wimmer and Lewis, 2010), reciprocity (Block, 2015; Goodreau, Kitts, and Morris, 2009) and popularity effects. In conjunction, we account for when students may come into contact before or more frequently than within the university - such as knowing each other before beginning university or living with one another. These are based on name generator questions, “Whom did you know before the start of the degree?” and “Whom of your fellow students do you live with at the moment?” were students name their prior and current relations. These are subsequently actioned as dyadic covariates in the models to account for friendship processes outside the university.

3.3. Methodology

We analysed the data using Stochastic Actor Oriented Models (SAOM) from the R package RSiena (Snijders et al. 2024). SAOM is a statistical modelling technique that enables researchers to examine social network dynamics, specifically the co-evolution of networks and actor characteristics over time (Snijders, 2005, 2017; Snijders et al. 2024; Snijders, Van De Bunt, and Steglich, 2010). As input, SAOM requires a series of network observations spread over time combined with relevant covariate data on the actor characteristics that might influence

the formation of social relationships between individuals – being our surface- and deep-level traits. SAOM models the change in the network as a result of a continuous process of tie creation, maintenance or disappearance resulting from actors’ decisions given the current network(s) state. These network changes are governed by functions which SAOM fits to the data, reflecting how well a modellers’ assumptions about the network-generating process allow to explain observed network dynamics. One of these functions is the so-called rate function, estimated from the data, determining how often an actor (or node) can change its ties between two waves. Further, the network change process is modelled as a sequence of events in which actors are randomly selected and then get the opportunity to create, retain or remove a single tie. The choice of this is made using the so-called objective function, with which the focal actor evaluates the consequences of a possible change along the dimensions that were specified by the modeller and reflect the preferences of actors specified by the modeller (e.g., on what characteristics they befriend others). This allows us to assess, for example, to which extent actors strive to form or maintain ties to demographically similar others or how important it is for them to reciprocate friendship nominations.

SAOM continues this simulation process, searching for the vector of model parameters that best fit the observed network dynamic so that the simulated network dynamic is similar enough to the empirical target network (Ripley et al. 2023). This is then used to interpret how important each of the effects specified by the modeller is for explaining observed network changes. A more technical outline of this SAOM process can be found in Snijders (2005) or the RSiena manual (Ripley et al. 2023). RSiena is valuable for this kind of analysis as it allows us to control for a range of external factors and processes that, while influential in forming or maintaining friendships, are already extensively studied. These include terms to control for reciprocity, the tendency for a respondent’s friends to name them as friends, and transitive triplets, the process of one’s friends also being friends with one another.

3.3.1. Time dummies

What differentiates each model in our study is the use of time dummies interacted with each homophily parameter individually, as seen in Table 3. This allows us to examine the effect of different surface- and deep-level characteristics on friendship individually. We examine the results stepwise, where the first model is a baseline model with no time dummies included, allowing us to assess whether our surface- and deep-level characteristics play a role in friendship at all (H1 and H2). All the models include all structural³ and control parameters and similarity on all surface- and deep-level characteristics, shown in S5 in the supplementary materials.

Subsequent models are analysed to see whether specific homophily effects become stronger or weaker over time. We specifically focus on similarity in selection, and, as a result, ego and alter effects are included in the model but not in the table below for brevity. We include these effects in the models to make sure that the estimate of the effect of two actors being similar in each characteristic (homophily) is controlled for simultaneous tendencies that actors with this characteristic nominate others more frequently or are nominated more frequently than those not having the characteristic. The similarity effects for social behaviour, partygoers, being serious (about studies), and being seen as a clever person are interacted individually with time dummies in separate models, with all else held constant.

3.3.2. Imputation

Network data often suffer from missing observations between waves due to student absenteeism or incomplete surveys. The Swiss Student

Life dataset is no exception, with up to 45 % of friendship nominations missing per wave, with data points not consistently coming from the same respondents across different waves.⁴ Maximum Likelihood (ML) multiple imputation methods, as proposed by Krause et al. (2018), are used to address this issue. These methods leverage information from previous waves to impute missing data, utilising both surface- and deep-level covariates. A detailed explanation of this procedure can be found in section 1.1 in the supplementary materials.

3.3.3. Homophily

To answer questions related to the macro effect of dyadic friendship choices, we use a well-established measure for observing within-group homophily – the Coleman index (Coleman, 1958). The Coleman index captures the degree to which the students in our network are more or less likely to create friendships with demographically similar others compared to a network of random mixing. The index also takes group sizes into account as larger groups have more chance to have in-group ties due to the abundance of similar others. A value of 0 indicates pure random mixing, negative values indicate heterophily, and positive values indicate homophily. By looking at this over each wave, not only are we able to compare how different groups are conducive to different homophily dynamics, but also how the overall levels of homophily within groups changes over the course of an academic year.

4. Results

4.1. Descriptive results

Descriptive statistics of the friendship networks are presented in Table 1, Table 2 and Fig. 1. The average degree ranges from 1.71 to 2.84 over the waves, which aligns with prior empirical studies on friendship (Berndt, 1982; Feld, 1991), albeit on the lower side. In general, females tend to get nominated and nominate more than males. For nationality, Swiss students tend to get nominated more than international students but nominate less friends. In all cases, the number of nominations increases up until the final wave, which is then slightly lower than wave 4. This is likely due to decreasing social interactions close to exam period at the end of the year. In terms of how much the network changes between waves, the Jaccard indices upward of 0.30 in Table 1 show that there was sufficient change in the friendship network over time to perform SAOMs (Ripley et al. 2023). This is also confirmed by the increasing number of ties found in Fig. 1. In general, the networks get more dense as time passes, with more ties created than terminated in each wave. Overall, the networks are as expected for effective SAOM models.

Table 1
Instance of tie changes and maintenance between the different measurement points.

	W1-2	W2-3	W3-4	W4-5
Tie Formation (0 → 1)	552	392	467	321
Tie termination (1 → 0)	213	330	328	306
Tie maintenance (1 → 1)	474	751	911	976
Jaccard Index	0.383	0.510	0.534	0.609

³ Specifically, we include Density, Reciprocity, Trids, In- and Out-group popularity including their square root terms, and an interaction between triadic closure and reciprocity as per Block (2015)

⁴ We evaluated the model’s fit by comparing the observed networks with simulated ones derived from a single imputed iteration of the baseline model in Table 3 (Lospinoso and Snijders 2019). More information on the goodness of fit can be found in S11 in the the supplementary materials

Table 2

Average in- & out-degree, over the waves and for subgroups (un-imputed data, with structurally absent respondents removed). Rows in italics are the average degree for the individuals that hold that specific traits. In a closed network, in-degree = out-degree and thus is only reported once.

	W1	W2	W3	W4	W5
Average in-/out-degree	1.71	2.15	2.47	2.84	2.39
in-degree					
By Gender					
<i>Male</i>	1.74	2.15	2.47	2.89	2.42
<i>Female</i>	2.41	3.11	3.65	3.77	3.35
By Nationality					
<i>Swiss</i>	1.89	2.30	2.68	3.11	2.57
<i>Non-Swiss</i>	1.68	2.22	2.52	2.73	2.48
out-degree					
By Gender					
<i>Male</i>	1.86	2.24	2.52	3.05	2.47
<i>Female</i>	2.39	3.57	4.38	4.23	4.15
By Nationality					
<i>Swiss</i>	1.99	2.40	2.75	3.15	2.63
<i>Non-Swiss</i>	1.74	2.48	2.86	3.35	2.88

5. Models

The models shown in [Table 3](#) and [Table 4](#) report only the homophily effects and their interaction with time dummies, which are our main theoretical interest. To explore different angles of deep level items, [Table 3](#) focuses on homophily regarding nomination-based interest traits, while [Table 4](#) features homophily based on self-reported personality traits. In addition, the models estimated contain network structural effects, controlling for tendencies to reciprocate tie nominations or complete friendship triads by new nominations. These effects are all significant and do not change substantively in subsequent models, which can be observed in the full [Table 3](#) and [Table 4](#) as well as S5 and S6 in the [supplementary materials](#). This is the same for control variables.

5.1. Baseline homophily effects

Model 1 in [Table 3](#) shows positive, statistically significant similarity coefficients for almost all homophily-driven covariates (SameX for binary traits (demographics) and SimX for non-binary traits (nominations and self-reports)). In the baseline model, being of the same gender, being both Swiss or non-Swiss, and whether students are perceived as similarly

clever, serious about studies, or partygoers are all important factors of friendship creation. This indicates that students prefer to make friends based on similarity on surface and deep-level characteristics, excluding being social or not, which is consistently non-significant. This aligns with an initial confirmation of H1 and H2, showing the presence of homophily within the cohort on almost all operationalised characteristics when considering all network waves.

In [Fig. 2](#), we can see a visualisation of Ego-Alter selection tables for our three significant deep-level characteristics of our baseline model in [Table 3](#). These graphs show the different possible numbers of nominations held by an alter (the lines separated by colour) and the contribution to the objective function (the Y-Axis) for each ego holding a specific nomination value (the X-axis). In other words, the higher the value on the Y-axis for a given curve, the more an actor holding a given characteristic with the corresponding X-value is inclined to select an actor with the value of the characteristic to which that curve belongs. More specifically, the values are the number of nominations a student got for each deep-level related trait, with all the waves and changing values considered. The results of this figure reflect what we see in our models - that homophily is present in most of our deep-level characteristics. For the different number of nominations held by the ego in the different lines in [Fig. 2](#), the contribution to the selection function is at its maximum at the values similar to the number of nominations held by the ego, leading to the curvilinear shape. For example, in all the significant deep-level traits, the pink and purple lines, representing those individuals who were nominated 9 and 10 times, respectively, are consistently increasing, being a maximum at the value of 9 and 10 on the X axis (the value of the ego).

The inverted U-shaped curve seen in many of the middle-nominated egos shows that they are most likely to befriend others who are also middle-level in how many times they were nominated (similar to them) and less likely to befriend those further away from their number of nominations. In addition, we can see that slopes are less steep at lower levels of ego nominations over all levels of alter nominations when compared to higher numbers of nominations. This indicates that those that are perceived as less party-going, serious about studies or clever are more indifferent about who they select as friends compared to those highly nominated students – who prioritize befriending other highly nominated individuals. Overall, for each number of nominations in our significant deep-level traits, the closer the alter is to the number of nominations held by the ego, the higher the contribution to the selection

Table 3

Stochastic Actor Oriented Models predicting friendship formation and maintenance driven by homophily and its change over time. The different models are unique in which covariate is interacted with time dummies.

	Model 1: Baseline	Model 2: Intl Student	Model 3: Sex	Model 3: Clever	Model 5: Serious	Model 6: Social	Model 7: Party
Intl Student Same	0.12** (0.03)	0.09** (0.04)	0.12*** (0.03)	0.12*** (0.03)	0.12*** (0.03)	0.12*** (0.03)	0.12*** (0.03)
Sex Same	0.19*** (0.04)	0.19*** (0.04)	0.16*** (0.04)	0.19*** (0.04)	0.19*** (0.04)	0.19*** (0.04)	0.19*** (0.04)
Clever Sim	0.47* (0.22)	0.46* (0.22)	0.48* (0.22)	0.45* (0.22)	0.45* (0.22)	0.47* (0.21)	0.45* (0.22)
Serious Sim	0.55** (0.17)	0.54** (0.17)	0.53** (0.17)	0.56** (0.17)	0.55** (0.17)	0.56** (0.17)	0.57*** (0.17)
Social Sim	0.10 (0.15)	0.09 (0.15)	0.09 (0.15)	0.10 (0.15)	0.09 (0.15)	0.09 (0.15)	0.11 (0.15)
Party Sim	0.85*** (0.17)	0.89*** (0.17)	0.90*** (0.17)	0.90*** (0.17)	0.89*** (0.17)	0.91*** (0.17)	0.89*** (0.17)
Time Dummy 2 * Sim/Same		−0.29*** (0.07)	−0.28*** (0.07)	0.02 (0.64)	−0.29 (0.51)	0.21 (0.40)	−0.25 (0.50)
Time Dummy 3 * Sim/Same		−0.18** (0.07)	−0.14* (0.07)	0.46 (0.57)	0.19 (0.45)	−0.10 (0.39)	−0.24 (0.50)
Time Dummy 4 * Sim/Same		−0.36*** (0.08)	−0.31*** (0.07)	−0.26 (0.60)	−0.14 (0.46)	−0.73 (0.46)	−1.25* (0.52)

Notes: Standard errors in parentheses. * + $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$;

All models include ego and alter effects, control variables, and structural effects. Full model results can be found in the [supplemental material](#). Sim(ilarity) and Same pertain to the homophily covariates (Sim for deep-level traits, Same for surface-level traits). The final three rows of each model show the interaction with time dummies

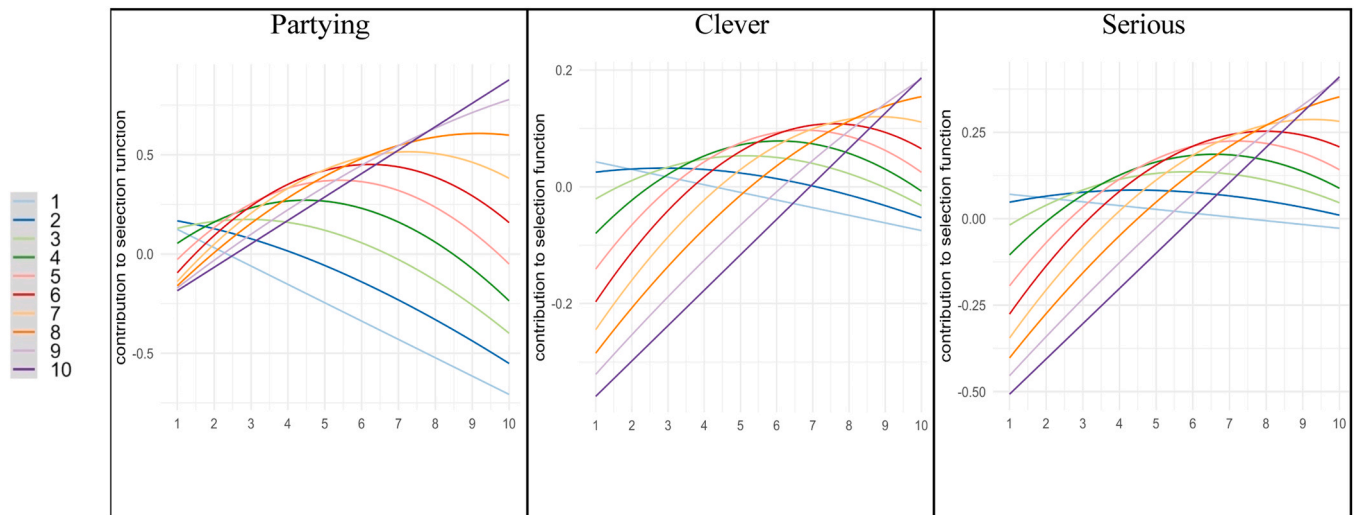


Fig. 2. Visualization of ego-alter selection tables for the significant deep-level traits used in our first series of models.

function, i.e. the more likely they are to be selected for friendship by the ego, assisting in confirming initially H1 and H2, without considering changes over time.

5.2. Dynamic homophily effects - nomination based models

Subsequent versions of our models all differ from one another in which homophily effect was interacted with time dummies – to observe the strength of these effects in the friendship dynamics at different periods while keeping all else constant. Models 2 and 3 in Table 3 show time-dummy interactions with international student status and gender homophily. Compared to wave 1, model 2 displays a significant negative effect at each subsequent wave of students with the same international student status on forming or maintaining friendships. This is the same for time-dummy interactions with gender showing significant negative effects. This indicates that while demographic friendship homophily is present in wave 1, its primarily driven by friendship in the first period.

So much so, that this homophily preference is negative in subsequent periods, indicating heterophily in peer relations in these periods. In other words, in later stages of familiarisation in the cohort, fewer friendship ties are being created and maintained along the lines of demographics than wave 1, which is in line with hypothesis H3a. It is also important to consider how this influences H1 – the overall level of surface-level homophily. While present in the models without time dummies, there is indication that, in some waves, there is a tendency to select demographically dissimilar others. Fig. 3 below as well as odds ratios in S10 in the appendix illustrate this decrease in homophily over time. This reduction is particularly visible among the male students whose homophily preferences drops below 0 in all waves after wave 1. Being the majority group, it is possibly a driving factor of our model effects. For the remaining demographic subgroups, while decreasing levels of homophily are present in subsequent waves, it is still present. Thus, we can continue to tentatively accept H1.

To investigate the potential compensating mechanisms of this shift,

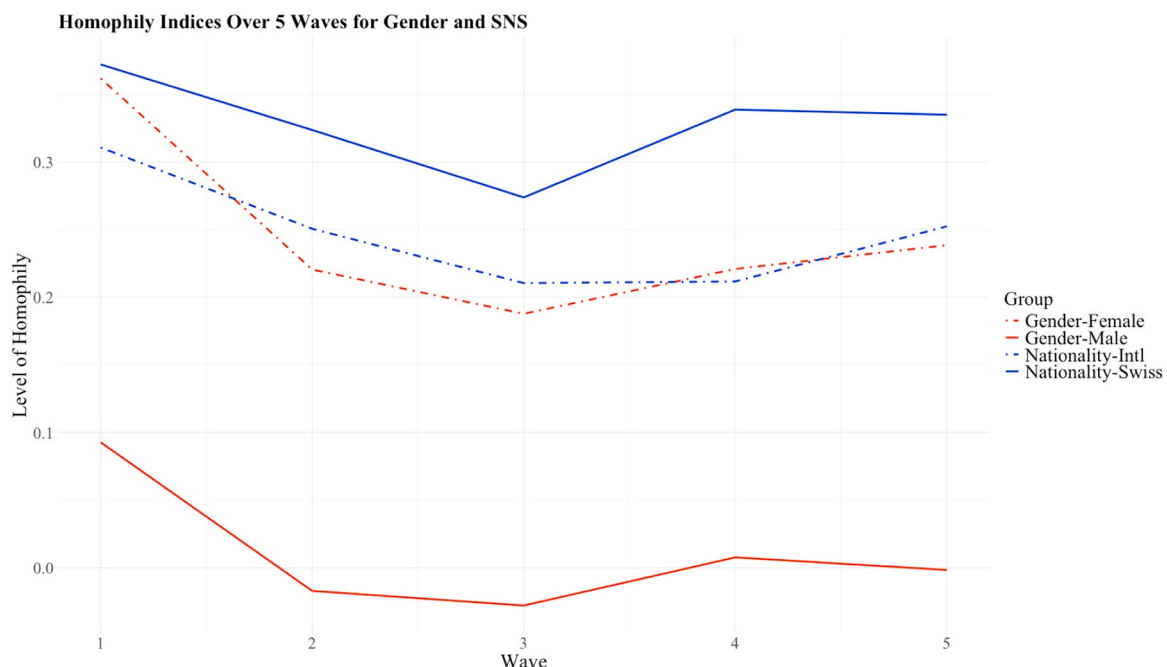


Fig. 3. Coleman homophily indices related to gender and international student status plotted over 5 waves of the student cohort.

we also interact a time-dummy with our deep-level traits: social, party, serious and clever. None of these interactions indicate a consistent significant increase or decrease in the likelihood of friendships being made along these deep-level characteristics compared to the first wave. This is backed up by weighted odds ratio plots in S10 in the appendix, illustrating very little change in within and between group ties when it comes to deep level traits. While there is a single negative effect in the final time dummy of partying, this is not enough to draw conclusions and is likely explained by the final wave being taken shortly before the end-of-year exam. As a result, there was less party-related activity. With this consistent lack of evidence for an increase of the homophily tendencies in these deep-level characteristics, we must reject hypothesis H3b.

5.3. Dynamic homophily effects – self-report models

While the results with indegree nominations for deep-level traits are consistent and relevant, questions can be raised about the use of indegree nominations to measure the similarity of deep-level characteristics in students. Students may know their friends better than other peers and thus primarily nominate the smartest/most dedicated/most social/partyingoers in their friend group. Thus, it is possible that these nominations will be updated in each survey wave to fit the students' current friend group, and thus, neither decrease nor increase their relevance in how they affect friendship over time. To further investigate this, we run more models exploring homophily on another set of self-reported deep-level traits available in the data. In particular, we tested for changing importance of homophily based on self-reported personality characteristics, more commonly called the Big-5 personality traits (Digman, 1990; McCrae and Costa, 1980, 1987). The five personality-related traits openness, conscientiousness, extraversion, agreeableness and neuroticism, fit our definition of deep-level traits as they are not easily recognisable before students know each other well enough.

We subsequently ran similar models to the nomination-based traits on each of the Big-5 traits to examine whether they follow similar trends (seen Table 4). As with nomination-based models, using baseline models

we see homophily on demographic characteristics, confirming H1. Similarity on some of the Big-5 traits are significant in the baseline model – namely, being similar to one another in extraversion and, to a lesser extent, conscientiousness. This is tentatively in line with hypothesis 2 – while there is homophily, it is only on a select few traits. For Agreeableness, which is often cited as a standard means for friendship to be based (Selfhout et al. 2010), we do not observe that similarity begets friendship. Like the nomination-based models, while there is a decrease in demographic homophily in each wave, there are no consistently positive significant interaction effects between time dummies and the homophily effects of our deep-level characteristics. In other words, growing homophily on Big-5 personality traits is not responsible for the lower surface-level homophily over time, confirming H3a but not H3b. It also shows that the lack of significant interactions with time dummies in our nomination-based models are not due to the reliance on peer nominations as similar results emerge for self-reported deep-level traits.

5.4. Effects on macro-level homophily

Our results indicate that demographic similarity becomes less important for friendship formation and maintenance over time. However, it remains an open question whether this translates into less clustering on demographic traits at the macro level between all groups. To test this, we plotted homophily indices for each demographic subgroup over time. Fig. 3 shows the Coleman Index (Coleman, 1958) for gender and international student status over the one-year period the cohort studies together. Fig. 3 shows that homophily on international student status and gender decreases after the first wave and remain at a lower level than wave 1, tentatively confirming our H3. Of particular interest is the majority groups' tendency to be heterophilic at some points. With the male group's homophily index being under 0 in waves 2 and 3 (and very slightly in wave 4 and 5), this proposes that over time, males very slightly prefer to befriend females than other males given the group sizes in the network. In general, it also seems to be the case that nationality has somewhat more of a homophilic element to it than gender. Overall, there seems to be a lower level of demographic

Table 4

Stochastic Actor Oriented Models on all covariate information, each model is different by individually varying the covariate interacting with a time dummy to see the effects of our covariates at specific points in friendship development.

	Model 1: Baseline	Model 2: Intl Student	Model 3: Sex	Model 4: Open	Model 5: Conc	Model 6: Extra	Model 7: Agree	Model 8: Neuro
Intl Student Same	0.11** (0.03)	0.08* (0.03)	0.11*** (0.03)	0.11*** (0.03)	0.11*** (0.03)	0.11*** (0.03)	0.11*** (0.03)	0.11*** (0.03)
Sex Same	0.20*** (0.04)	0.20*** (0.04)	0.18*** (0.04)	0.20*** (0.04)	0.20*** (0.04)	0.20*** (0.04)	0.20*** (0.04)	0.20*** (0.04)
Openness Sim	−0.12 (0.12)	−0.12 (0.12)	−0.12 (0.12)	−0.11 (0.12)	−0.12 (0.12)	−0.12 (0.12)	−0.12 (0.12)	−0.12 (0.12)
Conscientiousness Sim	0.28+ (0.16)	0.30+ (0.16)	0.31+ (0.16)	0.28+ (0.16)	0.29+ (0.16)	0.29+ (0.16)	0.28+ (0.16)	0.28+ (0.16)
Extraversion Sim	0.24* (0.12)	0.24* (0.12)	0.25* (0.12)	0.24* (0.12)	0.24* (0.12)	0.21+ (0.12)	0.23* (0.12)	0.24* (0.12)
Agreeableness Sim	−0.14 (0.13)	−0.14 (0.13)	−0.14 (0.12)	−0.15 (0.13)	−0.15 (0.13)	−0.15 (0.13)	−0.12 (0.13)	−0.14 (0.13)
Neuroticism Sim	0.15 (0.14)	0.15 (0.14)	0.15 (0.14)	0.17 (0.14)	0.17 (0.14)	0.17 (0.14)	0.17 (0.14)	0.15 (0.14)
Time Dummy 2 * Sim/ Same		−0.35*** (0.06)	−0.33*** (0.06)	0.10 (0.32)	0.08 (0.46)	−0.35 (0.33)	0.10 (0.42)	−0.43 (0.40)
Time Dummy 3 * Sim/ Same		−0.23*** (0.06)	−0.17** (0.06)	−0.42 (0.32)	−0.16 (0.44)	−0.87** (0.33)	0.72* (0.36)	−0.52 (0.39)
Time Dummy 4 * Sim/ Same		−0.36*** (0.07)	−0.30*** (0.07)	0.08 (0.36)	0.09 (0.48)	−0.69+ (0.38)	0.37 (0.37)	−0.27 (0.42)

Notes: Standard errors in parentheses. * + $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; All models include ego and alter effects, control variables, and structural effects. Full model results can be found in the supplemental material. Sim(ilarity) and Same pertain to the homophily covariates (Sim for deep-level traits, Same for surface-level traits). The final three rows of each model show the interaction with time dummies.

clustering among all groups in the network over time, albeit effecting females more than males while international student status is relatively similar between the two groups.

Using S7 in the [supplementary materials](#), we can also compare these clustering indices in our deep-level characteristics in [Table 3](#). Here, we separated our sample for each deep-level variable into groups based on the number of nominations received and plotted the change of the three clusters per variable over the 5 waves. These analyses show that for each construct, the students with multiple nominations are consistently clustered in a separate component away from the rest of the sample, hovering above an index value of 0.1–0.2. In other terms, the highest nominated individuals are consistently more well connected to other high performers than to the other lesser nominated students, who hover around or below zero, implying random mixing. This indicates that the lowest nominated students do not form clusters based on the perceived deep-level traits of others. The stable grouping of the top nominated students in our deep-level characteristics while demographic clustering decreases over time ([Fig. 3](#)) shows that time plays a role in integration among demographic characteristics but not among deep-level characteristics. This corroborates the SAOM results showing no significant change in homophily based on deep-level characteristics over time.

6. Discussion and conclusions

We aimed to further academic understanding of the role homophily in newly forming social networks from a dynamic perspective, differentiating between homophily on readily observable ‘surface-level traits’ like demographic characteristics and homophily on more hidden ‘deep-level traits’ like values and interests. Our empirical results corroborate previous research showing demographic homophily on gender and nationality ([Ibarra, 1995](#); [Kalmijn, 2002](#); [Leszczensky and Pink, 2015, 2019](#)), as well as differing homophily dynamics for different demographic groups ([Goel et al. 2021](#); [Pignolet et al. 2024](#); [Sajjadi et al., 2024](#)). In conjunction, we found that the importance of different forms of homophily changes over time. Specifically, we observed that gender and nationality homophily became less important as time passed. This contrasts with homophily on both self-reported as well as nomination-based deep-level characteristics which, while also present in friendship formation and maintenance early on, was consistent in its importance over time. The impact of similarity in being an extravert, conscientious, enjoying partying, being clever, and being a dedicated student appeared to neither increase nor decrease over time. This contributes to homophily literature by demonstrating that there is a temporal aspect in friendship preference and homophily on different characteristics follows different trajectories over time.

According to the by-product hypothesis ([S. Smith et al. 2014](#)), demographic homophily is rooted in the idea that demographic traits signal similarity in cultural values and interests. Selecting demographically similar friends may thus lead to friends who are similar on relevant deep-level characteristics too. However, the correlation between demographics and interests is far from perfect ([Stark and Flache, 2012](#)). Overtime, individuals may discover that they are not as similar as assumed, and once more is learned about others in the cohort, these demographic similarities become less relevant. Because the cohort of students considered in our study jointly attended the same classes and introduction events, they could efficiently learn about similarities of others that are not only based on demographics but also on deep-level characteristics that may be more important for friendship choices ([Byrne, 1997](#)). As a result, we saw the quickly eroding reliance on demographic characteristics as a reason for forming and maintaining friendships.

While we did observe homophily on deep-level characteristics, its relevance for friendship formation did not increase over time. This implies, that the deep-level traits included in our study are consistently influencing friendship formation and maintenance over most waves. By comparing the decrease in surface-level homophily over time to our

finding of constant deep-level homophily, we add weight to [Lazarsfeld and Merton's \(1954\)](#) argument that friendships based on values and interests may be more robust and long-term than those based on surface-level traits. Future research should continue our exploration with “deeper” deep-level characteristics. While partying and studying or being extroverted are very salient in study contexts where students are balancing student life, such characteristics could be even further decomposed into underlying traits. There are opportunities to look at specifics such as music genres, favourite subjects, more granular hobbies and so forth that can get deeper and more detailed in the similarities people have, see [Beek and colleagues](#) for an example ([Beek et al. 2024](#)).

How can we understand the absence of the hypothesized increasing importance of deep-level homophily over time? One possible explanation could be that the process of becoming familiar with each other happened significantly quicker than assumed. With integration events in the first month of studies, finding those like oneself in deep-level traits may have been easy for the students in our sample. Additional analyses showed that high clustering of the top number of nominated students for each deep-level trait was already present in wave one, suggesting the identification of similar others early on. This means a robust relationship is likely established and maintained from an early stage of the study programme. The use of a closed university cohort, all meeting at the same time and going through the same parallel process, may have allowed for a fast identification of these deep-level characteristics. Thus, it is important to extend this research into other contexts and with more information on meeting opportunities, which can add a more robust base to understand the underlying processes when people get to know each other.

Alternatively, once people found those similar enough to them in whatever characteristic they deemed important, they may have stopped searching for prospective friendships. With a lack of more specific events and contact data, we assumed that this closed cohort of students encountered one another through social and academic events, such as lectures or parties. This assumption is valid as, in other empirical work, friendships are often modelled through similarly exogenous random or neighbourhood driven processes ([Jackson and Rogers, 2007](#)). However, after an early phase of getting to know each other, this assumption could be incorrect: Once people found some friends that are similar enough on relevant characteristics, familiarization with the rest of the cohort may stop, explaining the lack of increasing relevance of deep-level characteristics. Addressing limitations on finding more detailed information about the possible explanations of this is encouraged for future papers. There is also the possibility that other relevant deep-level characteristics were not included in the data. In a cohort of hundreds of students, students may select friends based on similarity related to a significantly broader range of hobbies, interests, and personality traits ([Dalen and Seippel, 2021](#); [De Matos Fernandes et al., 2025](#); [Eklund and Roman, 2017](#); [McPherson et al. 2001](#)). Homophily on some of these dimensions may have increased, which could explain the lower level of demographic homophily, while the increasing importance of these deep-level characteristics was not observable in our study because we did not have data on these dimensions. To address this limitation of our study, further investigations would be advised to explore homophily dynamics on more interests and other deep-level traits.

Another point to mention is the social selectivity of highly nominated students in their deep-level traits identified in the additional analyses. With the early establishment of friendships along these lines in the first wave of observations and the absence of change of importance, we also found that the top nominated students possessing each deep-level trait are clustered separately from the other students. Our figures also indicate a selective preference for similarly highly nominated individuals. From this, we identified that a particular driving force of this deep-level homophily originate from the most highly nominated students for each construct. These groups for each deep-level traits were more selective in their choice for friends – overwhelmingly selecting those similar in level to them in the relevant trait. In contrast, the less nominated individuals

were less driven by selecting friends that are like them on the specific trait. These two findings indicate high “performing” clusters on these relevant dimensions. This is similar to Brouwer et al. (2018), who show that in an academic context, there is a well-connected cluster of high-performing students who prefer to interact with and seek help from one another. Our results suggest that this may also apply to non-academic characteristics warranting further inquiry into the extent of this generalizability, which further research can verify.

This paper sheds light on the dynamic nature of homophily, as well as the different sources of homophily. While surface-level homophily decreased over time, homophily on interests and personality traits remained constant, enhancing our understanding on the friendship formation and maintenance process among strangers. Further insight into which specific groups drive this process also expands on research on high “performing” clusters – and one can imagine this also holds for other deep-level characteristics that are not in the data. The implications for this are in line with what we expected – that people who identify with a certain trait or interest are driven to befriend and maintain these friendships with others who think similarly to them. Future research could explore self-reported level of identification with a broader range of deep-level interests and personality traits, which will offer a more robust base on understanding how and why individuals are friends with one another. Addressing the limitation of incomplete data, subsequent imputations and the effects on the goodness of fit (found in section 1.2 and S11 in the supplement) in this paper by using more complete and diverse friendship networks will be of great contribution to this interesting and important domain of research. This could also lead to studies on interventions that make relevant deep-level characteristics salient in contexts where there is a need for cross-demographic interactions. In conjunction, future research into deep-level characteristics which may increase in the face of the decreasing demographic homophily, and shorter timeframes of analysis to observe the speed of this process will assist in further understanding these dynamics.

CRedit authorship contribution statement

Andreas Flache: Writing – review & editing, Supervision, Project administration, Conceptualization. **Vincenz Frey:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Tobias Stark:** Writing – review & editing, Validation, Supervision, Conceptualization. **Daniel Robert Cowen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

We have no known conflict of interest to disclose.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.socnet.2025.10.001](https://doi.org/10.1016/j.socnet.2025.10.001).

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